

Setting Up MongoDB Cloud Account



Estimated time needed: 15 minutes

Objectives

After completing this lab, you will be able to:

- Create an account on MongoDB Cloud
- Provision MongoDB Atlas database on MongoDB Cloud
- Create API key and secret to use in client code

Running the lab

This is an instructional lab that does not require a Skills Network lab environment. You will follow this lab on MongoDB Cloud.

Using MongoDB Atlas on MongoDB Cloud

You can [sign up for a free MongoDB Atlas cloud account](#). Authoring this lab does not require a credit card.

Once you have signed up, you can create a new project and a cluster using the [steps listed in the documentation](#).

Use the steps here for a quick reference, but you should follow the latest MongoDB documentation for the most up-to-date instructions.

1. Once you log in, you will create a new project and give it an appropriate name.



Cloudapp's ...



Access Manager

Billing

All Clusters

ORGANIZATION

Projects

Alerts 0

Activity Feed

Settings

Integrations

Access Manager

Billing

Support

Live Migration

CLOUDAPP'S ORG - 2024-04-06

Projects

Find a project...



Project Name

Database Deployments

Tags



Users

Teams

Alerts

ORGANIZATION

Projects

Alerts 0

Activity Feed

Settings

Integrations

Access Manager

Billing

Support

Live Migration

CLOUDAPP'S ORG - 2024-04-06 > PROJECTS

Create a Project

Name Your Project

Add Members

Name Your Project

Project names have to be unique within the organization (and other restrictions).

VectorDB Project

Add Tags (Optional)

Use tags to efficiently label and categorize your projects. A project can have a maximum of 50 tags. You can modify tags for the project later. [Learn more](#)

Key	Value	Actions
Select a key or enter your own	: Select a value or enter your own	
+ Add tag		
		0 TAGS

Cancel

Next

2. Leave the add members step empty and click Create Project.

The screenshot shows the Atlas Cloudapp's interface. The top navigation bar includes the Atlas logo, a dropdown menu for 'Cloudapp's ...', a settings gear icon, 'Access Manager', 'Billing', and 'All Clus'. The left sidebar lists 'ORGANIZATION' with sub-items: 'Projects' (highlighted), 'Alerts 0', 'Activity Feed', 'Settings', 'Integrations', 'Access Manager', 'Billing', 'Support', and 'Live Migration'. The main content area is titled 'CLOUDAPP'S ORG - 2024-04-06 > PROJECTS' and 'Create a Project'. It features a progress bar with 'Name Your Project' (checked) and 'Add Members'. Below this is the 'Add Members and Set Permissions' section, which includes a text input field for inviting users via email address. A list of members is shown with 'cloudappdevs@gmail.com (you)' as the 'Project Owner'. At the bottom, there are 'Back', 'Cancel', and 'Create Project' buttons. The 'Create Project' button is highlighted with a red rectangle and a hand cursor icon.

Atlas Cloudapp's ... Access Manager Billing All Clus

ORGANIZATION

Projects

Alerts 0

Activity Feed

Settings

Integrations

Access Manager

Billing

Support

Live Migration

CLOUDAPP'S ORG - 2024-04-06 > PROJECTS

Create a Project

✓ Name Your Project Add Members

Add Members and Set Permissions

Invite new or existing users via email address...

Give your members access permissions below.

cloudappdevs@gmail.com (you) Project Owner

Back Cancel Create Project

Project M

Project O
Has full a

Project C
Can upd

Project D
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data and
operation

Project D
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Project D
Can acce
indexes

3. The next step is to create a new Deployment or a Cluster. These terms are used interchangeably.

Overview

DEPLOYMENT

Database

Data Lake

SERVICES

Device & Edge Sync

Triggers

Data API

Data Federation

Atlas Search

Stream Processing

Migration

SECURITY

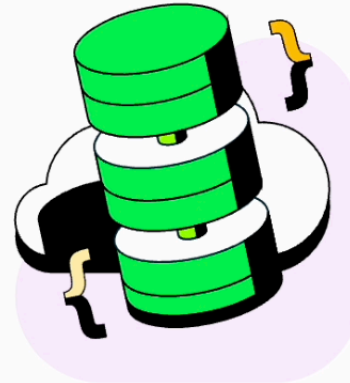
Backup

Database Access

Network Access

CLOUDAPP'S ORG - 2024-04-06 > VECTORDB PROJECT

Overview



Create a deployment

Choose your cloud provider, region, and specs.

+ Create

4. The following screen shows the different computer options. Ensure you pick m0, the free tier, for this lab. Give your cluster a helpful name and click Create Deployment. You can change the default provider and region if you like.

☐ M10

\$0.08/hour

For production applications with sophisticated workload requirements.

STORAGE

10 GB

RAM

2 GB

vCPU

2 vCPUs

☐ Serverless

\$0.10/1M reads

For application development and testing, or workloads with variable traffic.

STORAGE

Up to 1 TB

RAM

Auto-scale

vCPU

Auto-scale

☒ M0

For learning and exploring M environment.

STORAGE

512 MB

RAM

Shared



Free forever! Your M0 cluster is ideal for experimenting in a limited sandbox. You can upgrade to a production cluster anytime.

Name

You cannot change the name once the cluster is created.

vectordb-cluster



Automate security setup



Add sample dataset



Provider

aws



Google Cloud



Azure

Region



N. Virginia (us-east-1)



I'll do this later

Go to Advanced Configuration

5. MongoDB will create your username and password in the next step. You can change them if you like or use the default. Click Create Database User to finish this step. **Ensure you keep this username and password safe in a separate file or notepad, as you will need them for the lab's next steps.**

Atlas

Cloudapp's

VectorDB Project

Overview

DEPLOYMENT

Database

Data Lake

SERVICES

Device & Edge Sync

Triggers

Data API

Data Federation

Atlas Search

Stream Processing

Migration

SECURITY

Quickstart

Backup

D

N

Connect to vectordb-cluster

1

2

3

Set up connection security

Choose a connection method

Connect

You need to secure your MongoDB Atlas cluster before you can use it. Set which users and IP addresses can access your cluster now. [Read more](#)

1. Add a connection IP address

✓

Your current IP address (76.102.205.172) has been added to enable local connectivity.

2. Create a database user

This first user will have [atlasAdmin](#) permissions for this project. You'll need your database user's credentials in the next step.

We autogenerated a username and password. You can use this or create your own.

Username

Password

HIDE

Copy

Create Database User

Cancel

Choose a connection method

An IP address has been added to the IP Access List.

San

Click Choose a connection method to proceed to the next step.

Connect to vectordb-cluster

1 Set up connection security — 2 Choose a connection method — 3 Connect

You need to secure your MongoDB Atlas cluster before you can use it. Set which users and IP addresses can access your cluster now. [Read more](#)

- 1. Add a connection IP address**
 - ✓ Your current IP address (76.102.205.172) has been added to enable local connectivity.
- 2. Create a database user**
 - ✓ A database user has been added to this project.

You'll need your database user's credentials in the next step.

[Cancel](#) [Choose a connection method](#)

6. The next screen shows you how to connect to this database. You can explore more here or close the dialog.

Connect to your application



Drivers

Access your Atlas data using MongoDB's native drivers (e.g. Node.js, Go, etc.)



Access your data through tools



Data Explorer

Browse your Atlas collections without leaving the UI



Compass

Explore, modify, and visualize your data with MongoDB's GUI



Shell

Quickly add & update data using MongoDB's Javascript command-line interface



MongoDB for VS Code

Work with your data in MongoDB directly from your VS Code environment



Atlas SQL

Easily connect SQL tools to Atlas for data analysis and visualization



Go Back

Close

7. You just finished creating a project, cluster, and database. The final screen should show notifications confirming the same. You can click the Get Connection String to see how to connect to the database.

Overview

DEPLOYMENT

Database

Data Lake

SERVICES

Device & Edge Sync

Triggers

Data API

Data Federation

Atlas Search

S

Migration

Backup

D

N

A

CloudApp's Org - 2024-04-06 > VECTORDB PROJECT

Overview

Database Deployments

Create deployment ...

vectordb-cluster

Connect Upgrade cluster

Add Data →

Load Sample Data →

Data Modeling Templates →

Get connection string

A database user has been added to this project. ✕
Create another user in [Database Users](#).

An IP address has been added to the IP Access List. ✕
Add another address in the [IP Access List](#).

Deploying your data ✕
We are loading your new dataset to vectordb-cluster.

8. The system shows Node.js connection information by default, but you can change the driver to see other languages.

Connect to vectordb-cluster



Connecting with MongoDB Driver

1. Select your driver and version

We recommend installing and using the latest driver version.

Driver	Version
Node.js ▼	5.5 or later ▼

2. Install your driver

Run the following on the command line

```
npm install mongodb
```



[View MongoDB Node.js Driver installation instructions.](#)

3. Add your connection string into your application code

☐ View full code sample

```
mongodb+srv://cloudappdevs:<password>@vectordb-cluster.rjjdqf9.mongodb.net/?  
retryWrites=true&w=majority&appName=vectordb-cluster
```



Replace **<password>** with the password for the **cloudappdevs** user. Ensure any option params are [URL encoded](#)

Network Configuration

MongoDB Atlas only allows client connections to the cluster from entries in the project's IP access list. To connect, you must add an entry to the IP access list.

1. Open Network setting from your MongoDB Cloud account. You might see some IPs already inserted. If you see an IP of 0.0.0.0/0 in the list, you can skip the rest of the steps here.

+ ADD IP ADDRESS

You will only be able to connect to your cluster from the following list of IP Addresses:

IP Address	Comment	Status	Actions
146.70.174.186/32 (includes your current IP address)		● Active	⚙ EDIT 🗑 DELETE
0.0.0.0/0 (includes your current IP address)		● Active	⚙ EDIT 🗑 DELETE

If you don't see the IP 0.0.0.0/0 in the list, you need to add it by following the steps here. Click on [ADD IP ADDRESS](#)

Data Lake

 SERVICES

Device & Edge Sync

Triggers

Data API

Data Federation

Atlas Search

Stream Processing

Migration

 SECURITY

Backup

Database Access

Network Access

Advanced

IP Access List

Peering

Private Endpoint

+ ADD IP ADDRESS

You will only be able to connect to your cluster from the following list of IP Addresses:

IP Address	Comment	Status	Actions
146.70.174.186/32 (includes your current IP address)		 Active	 EDIT  DELETE

2. In the next screen, click on ALLOW ACCESS FROM ANYWHERE. This should populate the Access Entry List field with 0.0.0.0/0. Click on Confirm to close the dialog.

Add IP Access List Entry ×

Atlas only allows client connections to a cluster from entries in the project's IP Access List. Each entry should either be a single IP address or a CIDR-notated range of addresses. [Learn more](#).

ALLOW ACCESS FROM ANYWHERE

Access List Entry:

0.0.0.0/0

Comment:

Optional comment describing this entry



This entry is temporary and will be deleted in

6 hours ▼

Cancel

Confirm

Actions

⚙ EDIT

🗑 DELETE

your current IP address)

● Active

3. You should now see 0.0.0.0/0 added to the list of IPs in your Network settings.

IP Access List

Peering

Private Endpoint

+ ADD IP ADDRESS

You will only be able to connect to your cluster from the following list of IP Addresses:

IP Address	Comment	Status	Actions
146.70.174.186/32 (includes your current IP address)		Active	⚙ EDIT 🗑 DELETE
0.0.0.0/0 (includes your current IP address)		Active	⚙ EDIT 🗑 DELETE

You successfully set up a MongoDB cloud account and provisioned an Atlas database with the right network configuration. You can now use this account in the next set of labs.

Author

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